Design Write-up

Overview

Our system lets an expert remotely guide a novice through a 3D CAD part, teaching them which angles to look from and where the problems are. Each person sees the same part on their own table in augmented reality (AR) through their phones. The expert walks around the 3D CAD part and uses their phone as a tool to turn it, draw on it, pin problems with suggested fixes, and save viewpoints. Every action the expert takes appears live on the novice's side, and the novice can physically step into the expert's saved viewpoints.

System Definition

The system runs as a WebXR web app in Chrome on two Android smartphones, with partial iPhone support through Needle Go App Clip. It is used by one expert and one novice, synchronously and remotely, where they may talk through an external voice call while actively being involved in the same space.

The expertise conveyed is how to inspect a 2D part: where to look, which problems to point out, and how to fix them. Communication runs mainly from expert to novice, but the novice can reply by confirming pins or either through the external voice call. If needed, they could also request help from the expert with the "help" button where it will notify the expert of them being stuck.

Design Process

Initially, we diverged to brainstorm potential ideas of what the system might look like when focusing on expert-to-novice and novice-to-expert communication. We wanted to focus our specific attention on how to convey this information in a way that did not match up to traditional forms of remote and synchronous communication that already dominate as a way of cross-collaboration (e.g., video calling, voice calling, screen-sharing) while factoring in the expression of creative embodied inputs like gesturing, eye-tracking, or vocal cues to match to outputs of a similar nature. When actively going through the ideation process, we settled on an idea that we agreed upon that seemed interesting as a concept, which was the idea of having a mascot character representing an assistant to facilitate the communication pipeline between experts and novices when it comes to 3D art assisted modelling. With this direction, there were multiple ways to demonstrate individual and unique takes to how we could design this mascot figure involving its appearance as well as how it would respond to different cues provided by the expert and novice respectively in order to convey gestures as output (see *figure 1* and *figure 2* under *initial sketches*) for individual ideas, some of which include specific vocal cues mapped to essential actions like voicing “pinch” to have the mascot shrink in size or having the assistant see into what the novice is looking at to represent the expert’s perspective.

However, after deliberating as a team and factoring in for the time we were assigned to carry out the concept, we decided that due to technical feasibility, lack of general knowledge, and time constraints, we would not be able to implement this concept of having an AI-like assistant to direct actions and respond accordingly to all of our inputs. Therefore, we converged to ideate in a separate direction while still keeping the significance of non-traditional forms of co-creation in mind to highlight the importance of communication in a manner that would not suffer from the effects of screen-sharing or zoom calling (e.g., compression, lack of active presence and involvement, missing input and output vehicles that do not rely solely on voice calling with the feeling of sitting in separate rooms while staring at the same screen). From this, we were able to come together to form the concept of using AR to simulate the environment for embodied collaboration to happen.

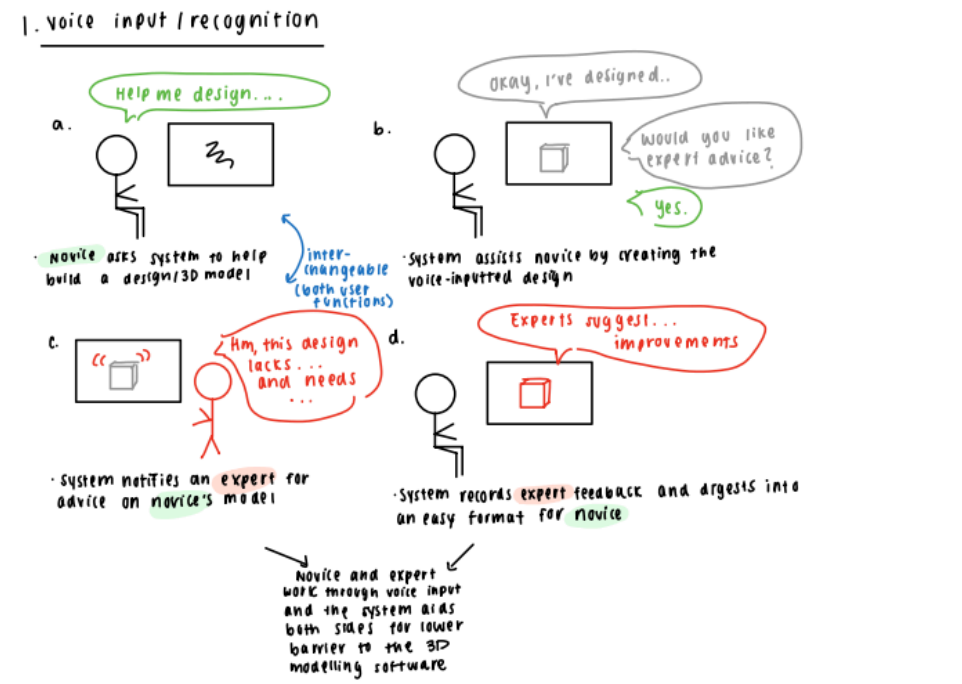
Without factoring in practical constraints in terms of doing conceptual sketches heading in this direction, we were able to come up with interesting and unique ways to carry out this sense of embodied interaction within an AR environment as we wanted to explore the concept due to its perceived creativity (see *figure 3* and *figure 4* under *concept sketches*) for ideas involving using a

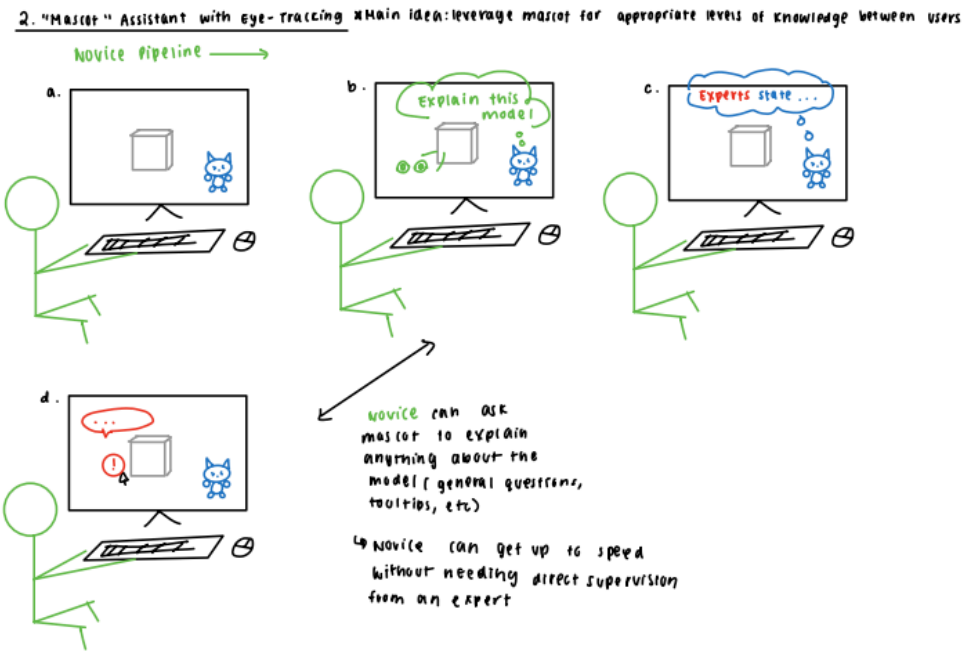
phone as a lens and controller to create a 3D space or as a hologram to an interactive canvas that would simulate telepresence as a more active form of digital art creation. Upon deliberating and discussing the potential ways of bringing this idea to life, we settled on using a phone as the means of communication to implement this concept in a realistic and time-sensitive manner in order to bridge communication in a bidirectional way between the expert and the novice.

Following this, we converged to figure out how to best implement this concept to an actual application in evaluating a design direction and which features would be involved to simulate a similar enough experience into the vision we had approaching the project. Generally, design decisions that we shared when factoring in interactions with the mobile device involved spatial feedback (e.g., rotation to move objects, movement with the phone, motioning to form a line projectile sketch, etc…) while creating a sense of a shared 3D space, which mimics presence over passive feedback. In terms of actual implementation, we settled on using WebXR as a way to mimic the 3D environment with phones being “ghost lenses” or a way of entry into these various actions as we mapped them onto buttons (see *figure 5* under *final sketch*). As suggested, these phone buttons would perform various actions assisting with 3D design such as placing CAD pieces, moving them, and placing pins to assist with guidance and feedback. To further refine our application, we added sounds to actions as well as haptics of different frequencies that would aid in assistance and ease of communication between the expert and the novice. While these features were mostly centered and built for expert-to-novice communication as we believe more input is necessary for guidance rather than receiving feedback, we also incorporated the ability for the novice to signal help from the expert if they needed to for more functional cross-communication purposes.

Figures

Initial Sketches





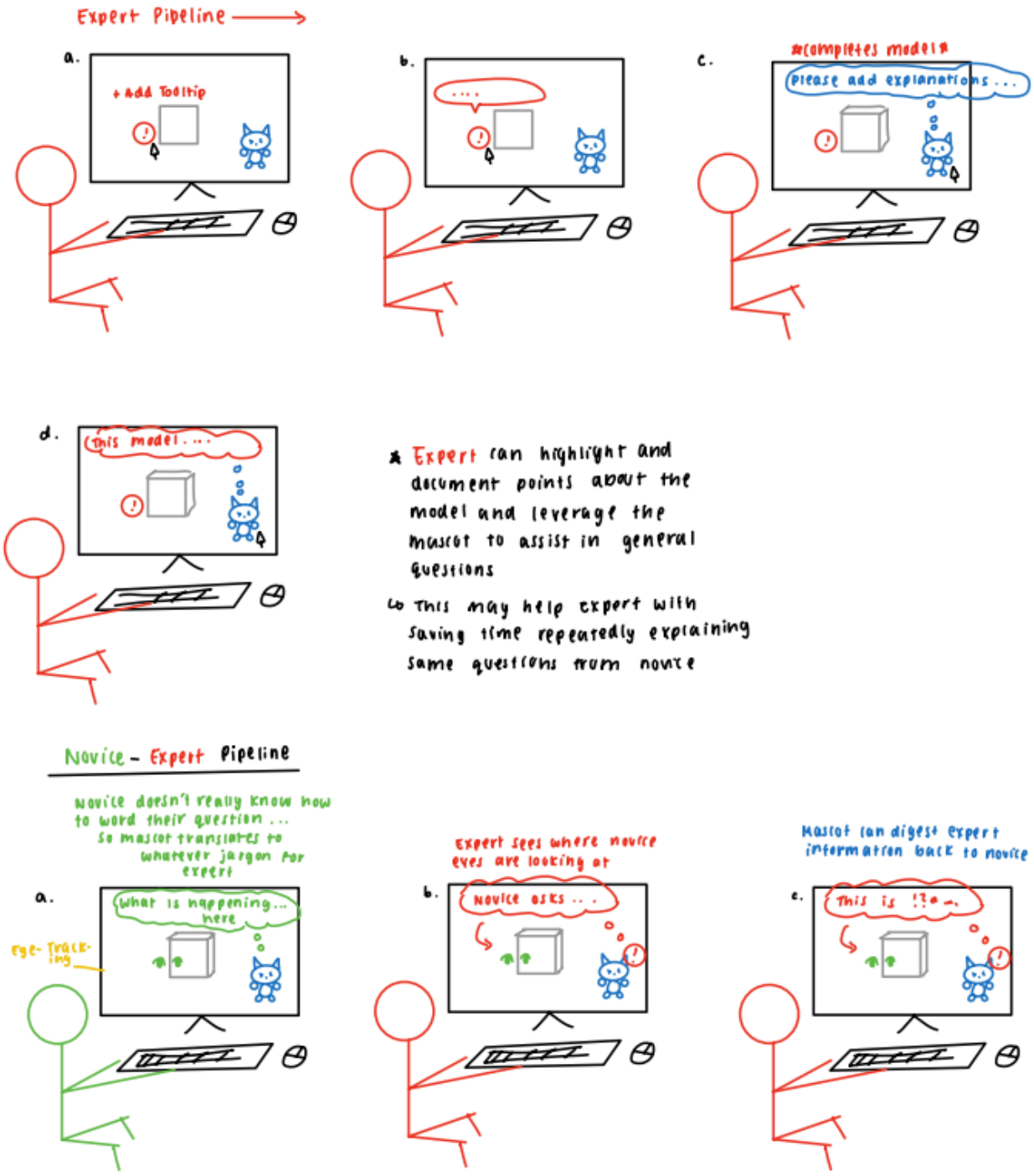
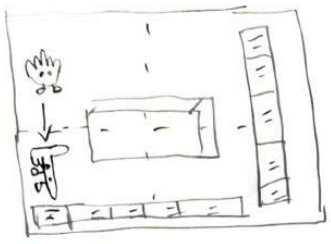


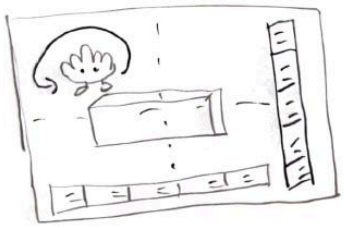
figure 1: Fion's initial brainstorming sketch featuring input and output communication

① Expert → Novice (INPUT: voice cues → OUTPUT: hand gestures)

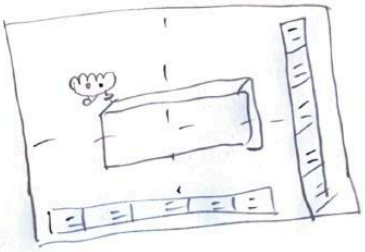
1. "look at [region]" → hand goes there + points to region



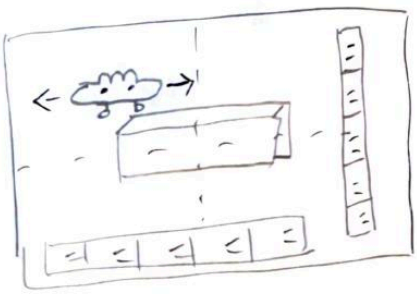
2. "Rotate" → hand turns around 360°



3. "Pinch" → hand closes into fist

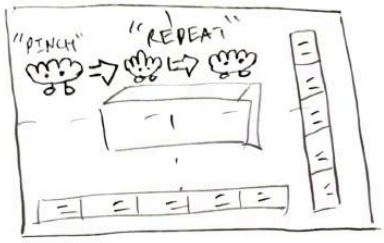


4. "Extrude" → hand extends into flat shape / becomes flattened



(2) Novice → Expert (INPUT: voice cues → OUTPUT: hand gestures)

1. "Repeat" → hand will do last action again (IDLE stake is "pen palm")



2. "Pause" → hand will freeze in last pose stake until next cue

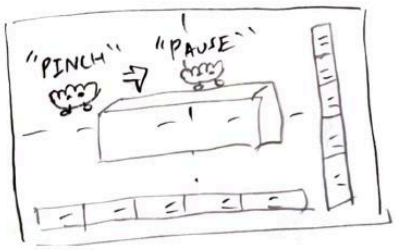


figure 2: Wish's initial brainstorming sketch featuring input audio and output cues

Concept Sketches

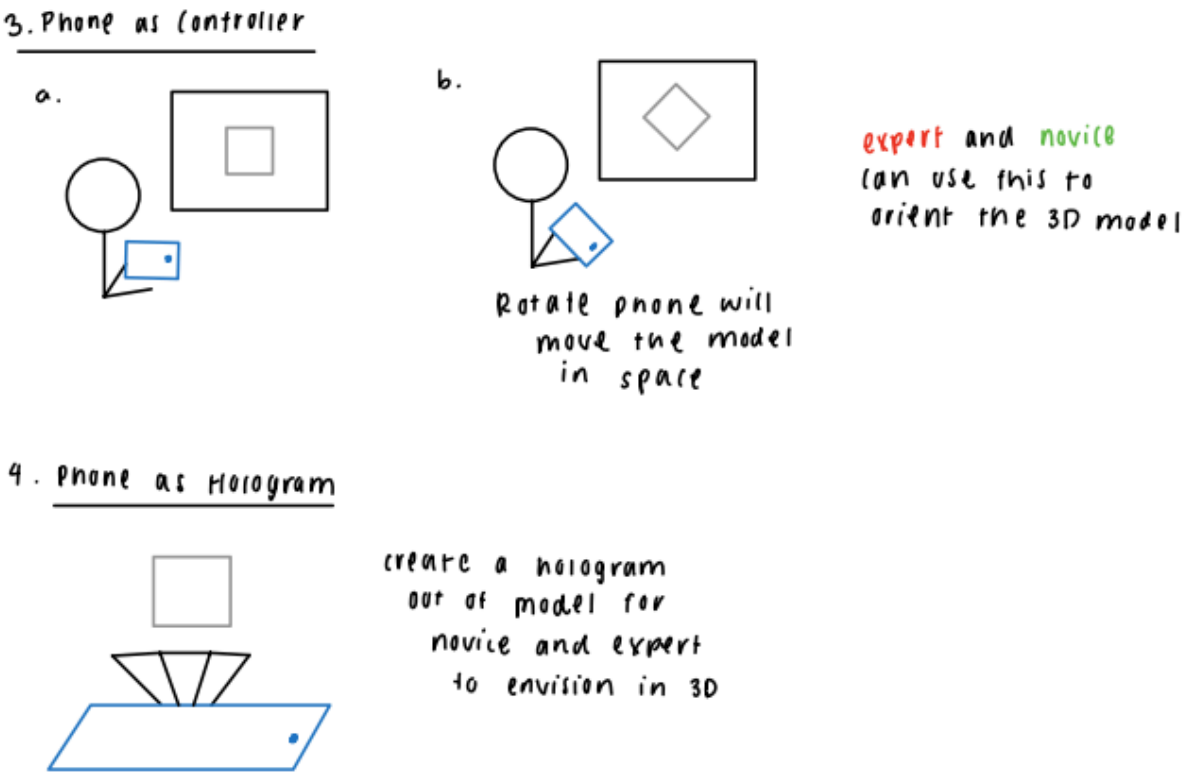
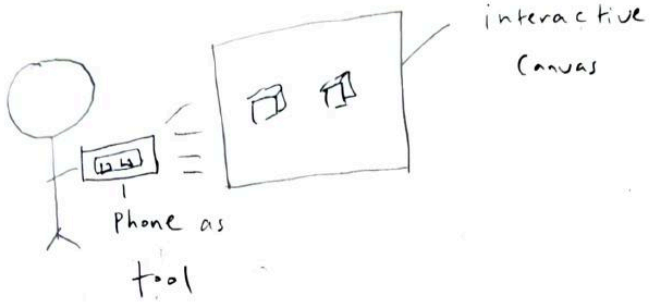


figure 3: Fion's concept sketch of ideas with phone movements and hologram modelling

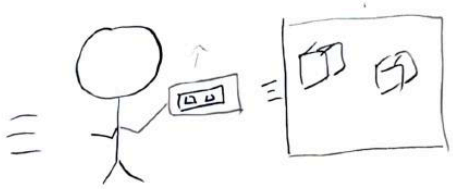
* Concept: 3D interactive space for art co-collaboration



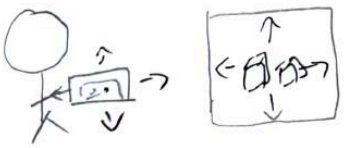
* From an expert or novice's perspective, they project object view out from phone to interact in a 3D space simulation

↳ Telepresence over traditional screen-sharing/calling (felt presence)

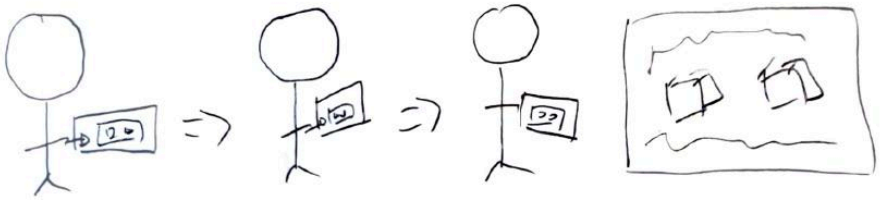
* Embodied inputs:



* Walking to tilt view (distance)



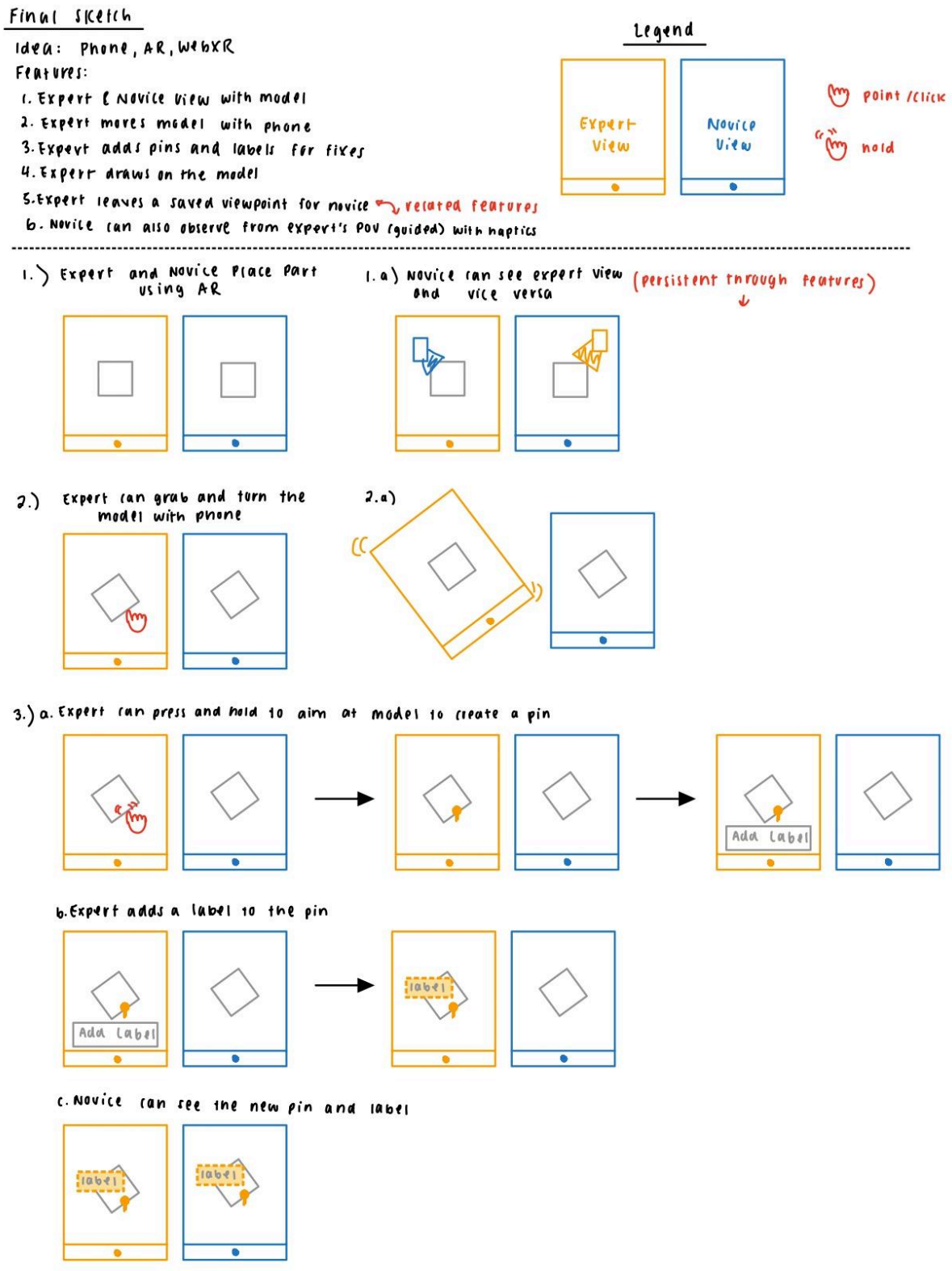
* Rotating will control projection direction (camera angle)



* Pressing down and moving phone will sketch up + down line projectile (e.g., magic wand waving in the air)

figure 4: Wish's concept sketch illustrating an interactable 3D canvas space with embodied input

Final Sketch



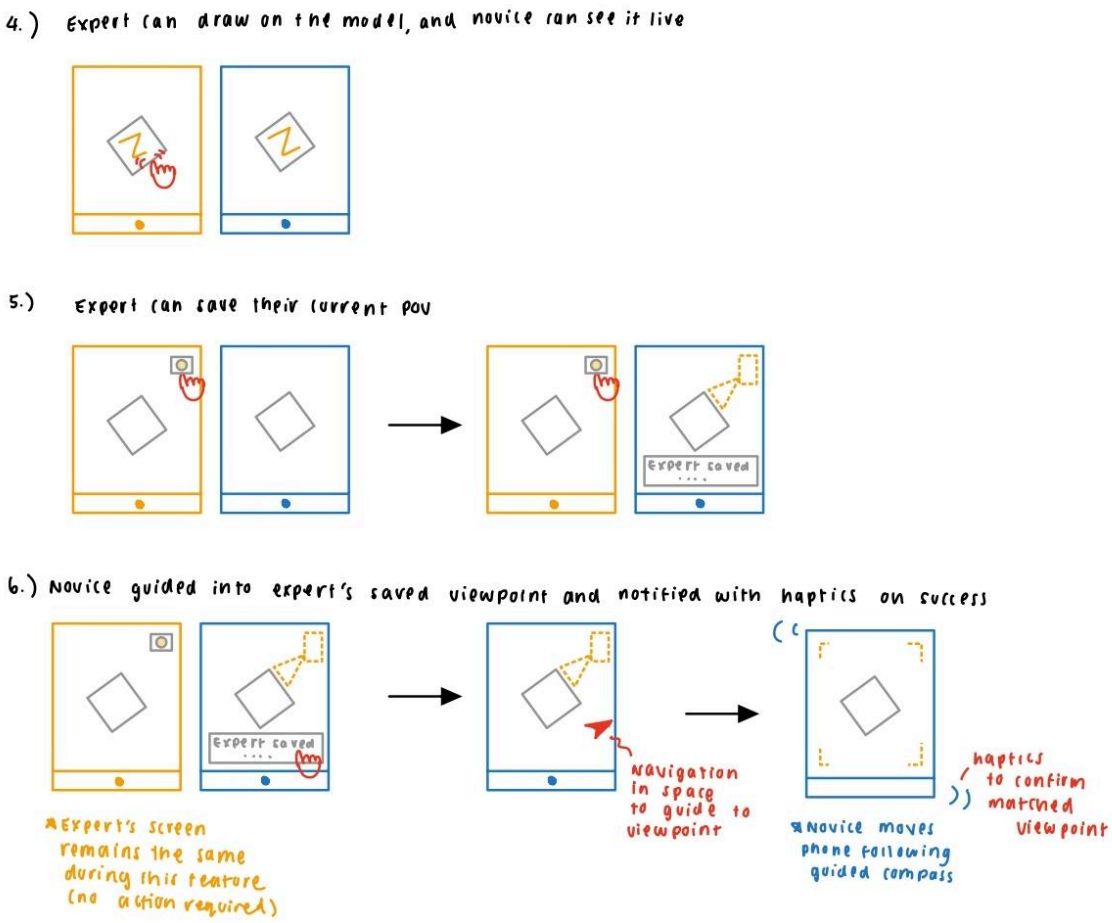


figure 5: Combined concept sketch picturing our implementation direction

Functionality

Usage

Listed below are the technologies used for our system:

Area	Tool
3D and AR	three.js & WebXR
Phone-to-phone sync	PeerJS
Build	Vite
Sound	Web Audio API
Hosting	GitHub Pages
iPhone AR Support	Needle Go App Clip

Limitations

Listed below are the current limitations of our system given resource and time constraints at the time of submission:

- **Cross-device incompatibility**: WebXR is not compatible with iPhone web browsing as it uses Safari so the Needle Engine is required for viewing purposes on iOS devices
- **Lack of vibration feedback**: iPhone lacks vibration support as WebXR was built for Android, therefore only sound is available when viewing on iOS devices which limits feedback on iOS devices

Future Implementations

Listed below are pieces of functionality that we would expand and design for if we were to continue development of the application:

- **Guided tutorial on how to use features:** an onboarding process or other means of help to introduce the users on how to best access and
- **Multiple experts and novices within the same room:** our application only supports the one-to-one connection of an expert and novice working within the same simulated space. However, expansion to facilitate more members and support multiple expert-novice connections within the same shared space would be beneficial for co-creation purposes

- **Custom CAD model uploads:** at our current stage, we have static and hardcoded models. However, if users could upload their own custom CAD models, it would create for a more dynamic experience
- **Integration into actual 3D software:** embedding the tool in an already existing CAD environment or software would provide a more realistic experience into what an everyday workflow would represent
- **Greater complexity of actions:** if the toolbar supported more actions and modeling operations beyond the basic ones implemented in our system, it could reflect a more expansive scope that shows up in real workflows
- **A wider variety of input to output feedback:** incorporation of other design elements that hold feedback (e.g., voice cue) could be helpful to support further assistance between novice and expert communication
- **Expansion on drawing features:** different drawing features added (e.g., different colours, lines, thicknesses, etc…) for the tool could help with more responsive communication between different parties
- **CAD fix previews:** feature to show the suggested fix on the part itself, so the novice can see the result before applying it, which would help visibility for the expert's guidance
- **Editable pin fixes:** ability to change the problem or fix from the pin drawer without removing and adding a pin again would provide for a more efficient experience in managing operations
- **Save and replay sessions:** allowing a novice to save and revisit an expert's pins, fixes, drawings, and viewpoints later could help with memory retention and walkthrough assistance
- **Add settings page:** an added settings page would allow the user to customize certain features (e.g. hide novice or expert's live point-of-view) to curate a more personalized viewing experience

Disclosure

Ideation and Design

We used Generative AI to refine our concept, create mockups, plan the work, and implement it.

- **Refining the concept (Claude Opus 5.5, chat):** Claude critiqued our brainstormed concept against the requirements and rubric and suggested technologies (WebXR, PeerJS, Needle's iPhone App Clip) and features. Example prompts used were:

- **Creating the mockup (Claude Opus 5.5, Design canvas):** Claude produced the storyboard of each scene and detailed views of the features (pins, pin HUD/drawer, fix labels, novice confirmation, and expert HUD layout). Example prompts include: "Create a mockup view of how this idea is being fleshed out so it is aligned with how I am imagining it with the refined concept", "Create a detailed mockup of the laser pin feature, and how the pins would look in 3D". We reviewed every mockup and iterated on changes to reflect our concept.

- **Creating the issues (Claude Opus 5.5, chat):** Based on the refined concept and mockup, we asked Claude to break down the features into GitHub issues that reflect the build order. Example prompt: "Create issues that build on the idea iteratively and descriptions of what needs to be done in order of implementation, according to the mockup, that will allow group members to work in parallel". We reordered them so the core features came first, then created and edited the issues in GitHub ourselves.

Implementation

Fion used Claude Code to implement the issues she worked on, one issue at a time. For each issue she asked it to plan only that issue before coding, review the plan, implement it, and review the changes. She approved each plan, reviewed every change, and fixed what it got wrong, and tested on phone/desktop before merging. Example prompts include:

```
- Read CLAUDE.md, docs/CONCEPT.md and docs/issues/NN-*.md. Plan how to
implement issue NN only. List the files you'll create or change, and
anything in the issue that's unclear. Don't write code yet. Follow
docs/DESIGN.md for every visual element. Point out anything the spec
doesn't cover instead of inventing a style.
```

```
- Review the uncommitted changes (git diff) for issue NN against
  docs/issues/NN-*.md, CLAUDE.md and docs/DESIGN.md. Check: Function
  - Every task and the "Done when" are met, and nothing from later issues
  was added
  - Shared data uses modelRoot local space, never world space
  - Synced transforms go on modelPivot; ghosts, strokes and views are
  children of modelRoot
  - Messages use the agreed types and stay small; pose updates are
  throttled
  - The desktop view still works
  - Bugs, leftover debug code, unused code, and missing error handling
  (e.g. AR not supported, connection lost)
- Design (docs/DESIGN.md)
  - Only the CSS tokens in src/style.css are used: no hardcoded colours,
  fonts or sizes
  - Orange only for the expert, blue only for the novice, never decorative
  - Layout: chips top-left/top-right, mode bar or toast at the bottom,
  nothing in the middle of the screen except the crosshair
  - Components match the spec (chip, mode button, toast, progress bar,
  scene label, crosshair); touch targets ≥ 48 px
  - 3D materials match the spec (part edges and contact shadow, translucent
  ghosts with depthWrite off, fat-line strokes, pin pulse)
  - Motion: 150–250 ms ease-out, no bounces, only the allowed idle loops,
  prefers-reduced-motion respected
  - Copy: sentence case, plain verbs, no exclamation marks; uppercase only
  on mono chips and labels
  - None of the "Don't" items: no gradients, glassmorphism, emoji, alert()
  or default browser buttons
- List the problems by severity with file and line, split into Function and
  Design. Don't change any code yet.
```

Daniel utilized Claude (Sonnet 5.5) strictly as an implementation assistance not for the design creation. The AI usage was used for debugging and tracing code to identify and resolve bugs that had occurred. With this being one of the few times he's worked with javascript, this resulted in the use of AI to act as a virtual tutor to explain the more intricate implementations needed to create a more cohesive product.

Arshaum used ChatGPT, including Sol and Astra, to assist with this project's implementation. Sol supported environment setup, troubleshooting, terminal commands, and Git/GitHub workflows. Astra generated and revised patches to code he had generated, as well as providing

proper test cases to ensure implementation is working successfully. He applied the changes to the local repository, ran the tests and production build, and manually checked expert and novice interactions in desktop browser tabs. Arshaum submitted the implementation for teammate review and tested the subsequent revisions. Physical-device AR testing was separate from these desktop checks.

Wish used Claude Opus 5.5 within the Cursor agent to assist with the implementation of the project details, including aspects related to troubleshooting, breaking down requirements in what needed to be implemented, how to plan around the implementation, execution, and managing Git workflows within the context of resolving merge conflicts and assisted code revision. Changes were manually reviewed within the implementation window and testing on Android devices as well as desktop was performed manually.

Demo

The demo video was generated from our code with brag (https://github.com/latent-spaces/brag), an open-source skill that reads a project's code and writes a short video plan, which the Hyperframes video tool then renders. We enabled its AI voiceover, which uses the Kokoro text-to-speech model, and asked for narration with no music. We gave it the features to show and their order, and reviewed the plan before rendering. The prompt is as follows:

```
/brag --voice --tone "clear, calm, product demo for a university HCI
critique"

 Requirements for this video:
- 100-120 seconds, never over 2 minutes
- Showing the main features in this order (with expert and novice
view): join a room, place the part, ghost phones, grab and turn,
mid-air sketch, add a pin and pick the fix, novice confirms the pin,
viewpoint matching with alignment and haptics
- Narration only: no background music, no sound effects (unless it is
from the code directly)
- Base visuals and wording on docs/CONCEPT.md and docs/DESIGN.md
(orange = expert, blue = novice); don't show features that aren't in
the code
- Show me the plan and script before rendering
```